

EduEase AI – Updated Problem Statement & Agentic AI Solution

1. Problem Statement

Educational materials often contain complex language that is difficult for learners with different knowledge levels. EduEase AI simplifies study materials and generates personalized learning resources.

2. Objectives

- Simplify academic content.
- Generate summaries.
- Create MCQs and exam questions.
- Explain keywords.
- Answer questions from uploaded content.
- Support Beginner, Intermediate and Expert learner levels.
- Auto-detect learner level.
- Generate personalized study plans.
- Detect difficult concepts for focused revision.

3. Agentic AI Workflow

The application accepts PDF or text input, extracts the content, processes it using IBM watsonx.ai, and performs multiple independent AI tasks including simplification, summarization, quiz generation, study planning, keyword explanation, question answering, and difficult concept detection.

4. Technologies Used

- Python
- Streamlit
- IBM watsonx.ai
- PDF processing utilities
- GitHub
- Streamlit Community Cloud Deployment

5. Current Features

- PDF Upload & Text Input
- Content Extraction
- Simplify Content
- Generate Summary
- Generate MCQs
- Generate Exam Questions
- Explain Keywords
- Ask Questions from Content
- Auto Detect Learner Level
- Generate Study Plan
- Find Difficult Concepts
- Responsive Web Interface

6. Expected Benefits

Improves student understanding, saves revision time, provides personalized learning support, and enables interactive AI-powered education through a deployed web

application.

7. Future Enhancements

- Multilingual support
- Retrieval-Augmented Generation (RAG)
- Learning analytics dashboard
- Voice interaction
- Full IBM Granite model integration where required

8. Conclusion

EduEase AI successfully addresses the Course Content Simplification challenge by combining an agentic workflow with IBM watsonx.ai, providing multiple educational AI services through a single deployed Streamlit application.